

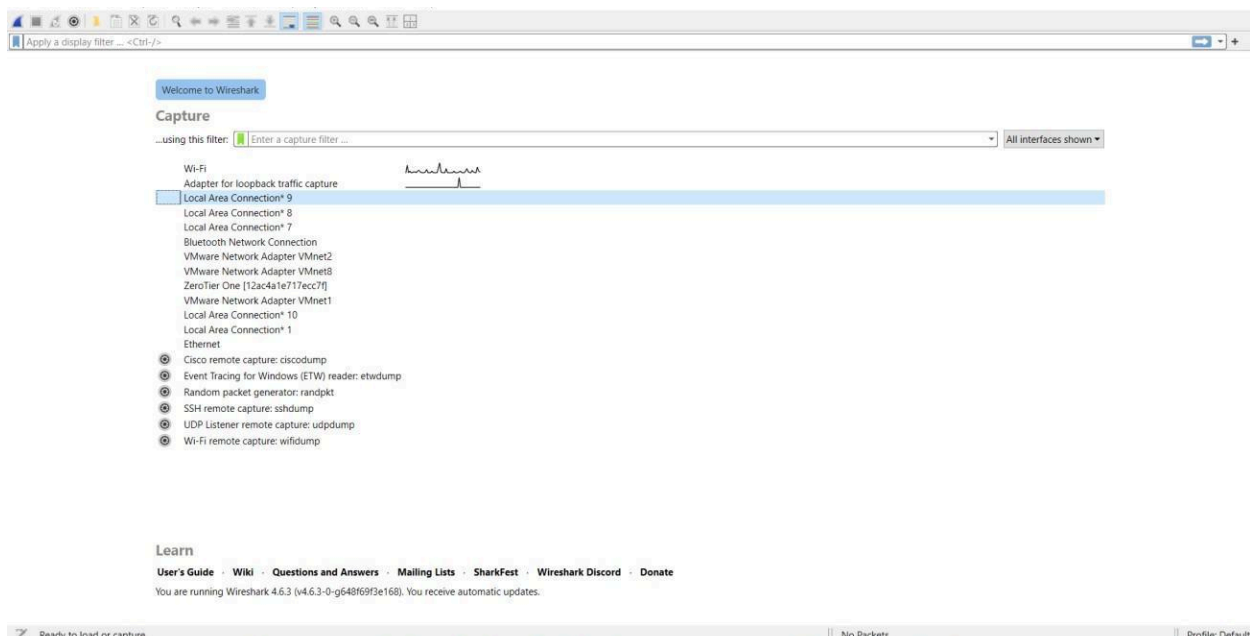
Advanced Computer Networks – PCS223

Assignment 2

Name – Taranpreet Singh

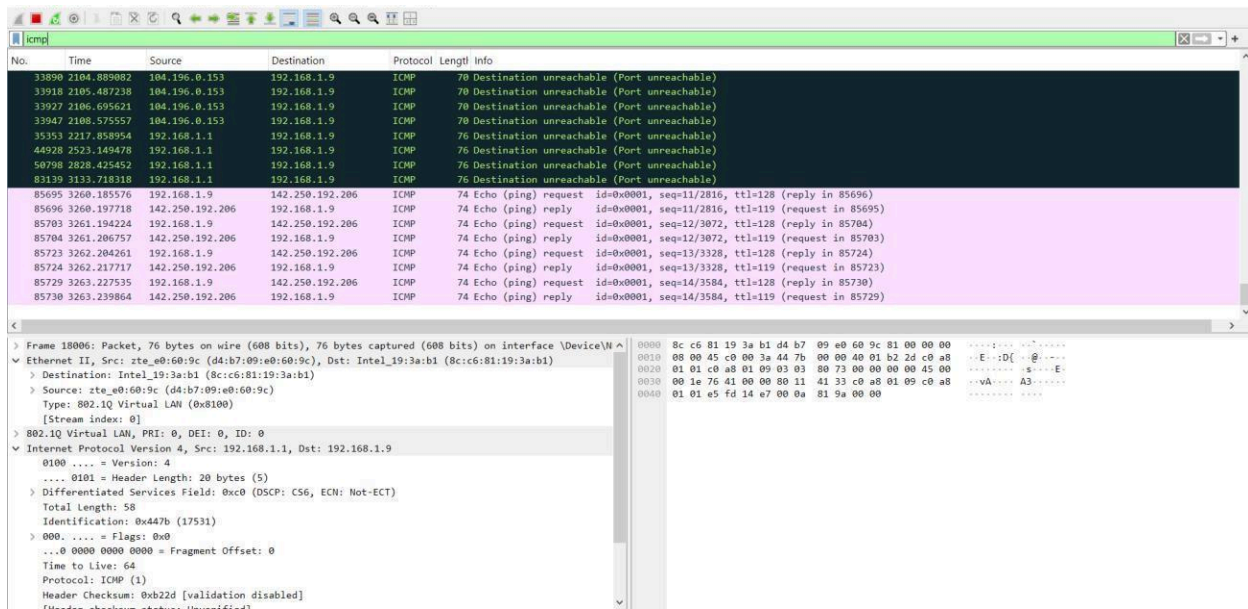
Subgroup – CS4

Roll No. – 8025320103

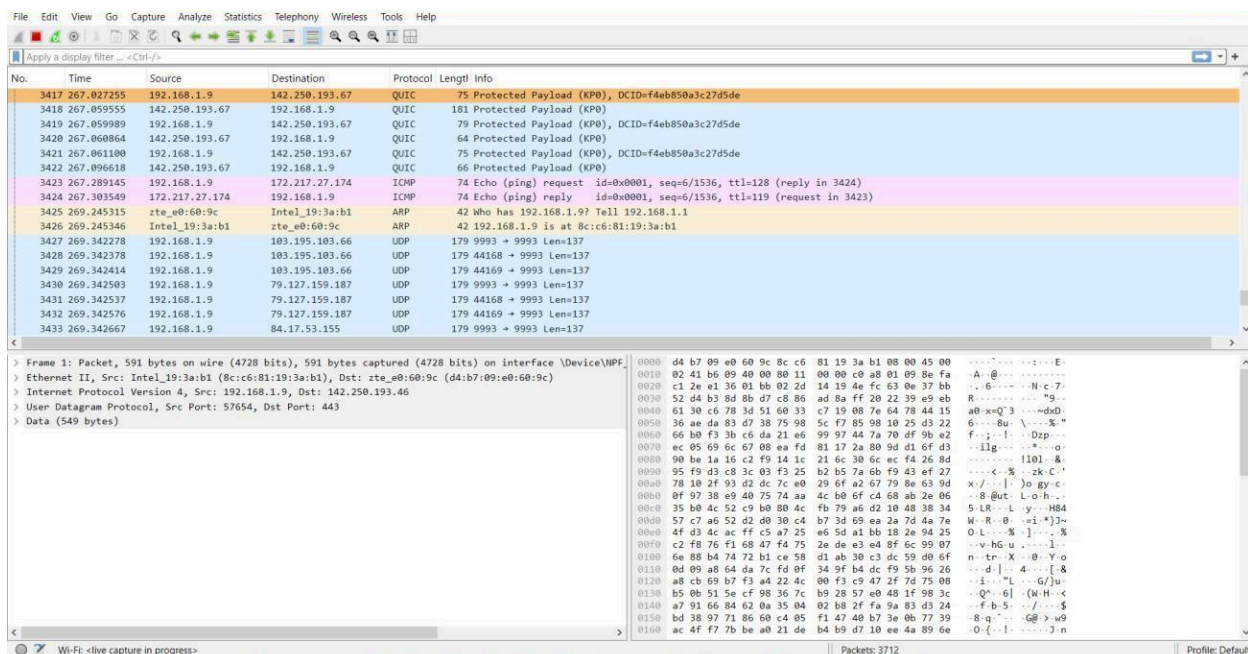


Screenshot showing Wireshark Home screen showing all interfaces and the active interface (Wi-Fi)

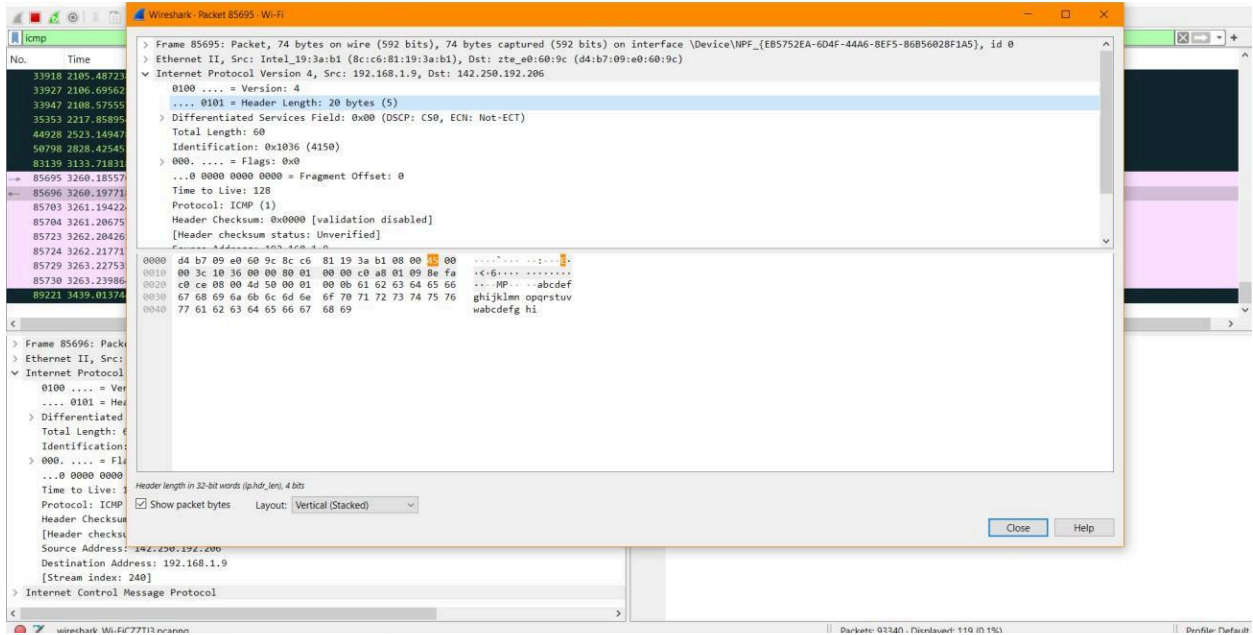
1. To capture and analyze ICMP packets (Ping request and reply)



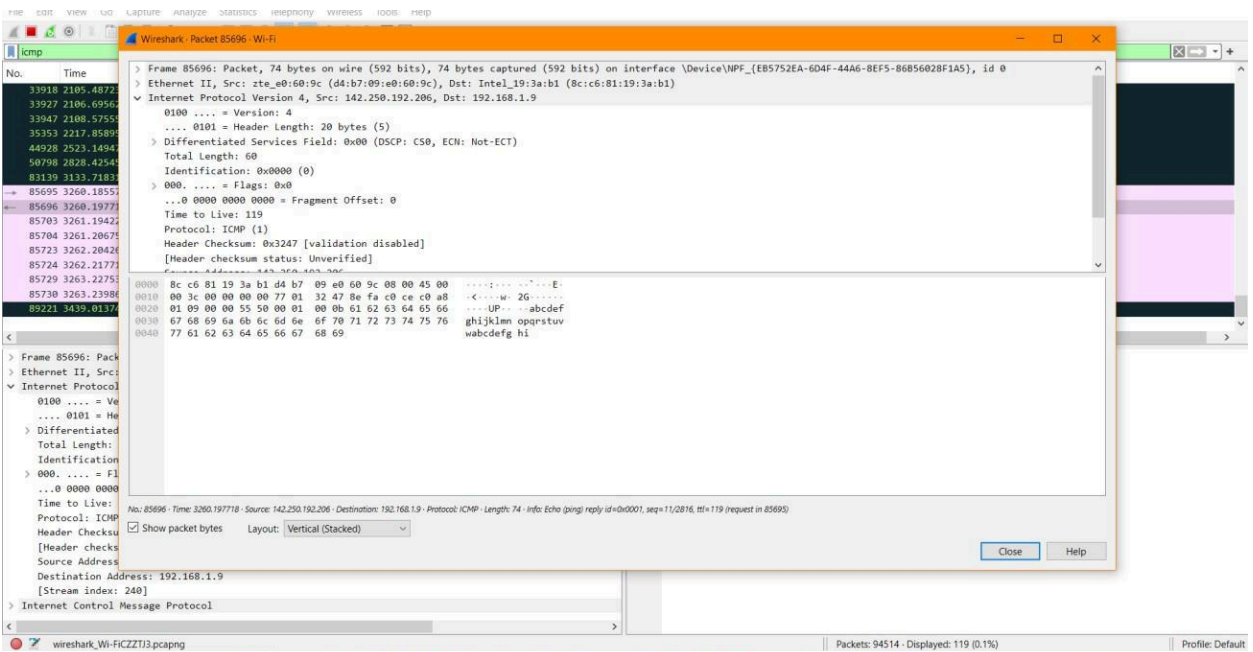
Screenshot showing ICMP Packets Captured by pinging google.com



Screenshot showing ICMP Packets Captured by pinging google.com



Screenshot showing ICMP Packet 85695 details (Ping Request)



Screenshot showing ICMP Packet 85696 details (Ping Reply)

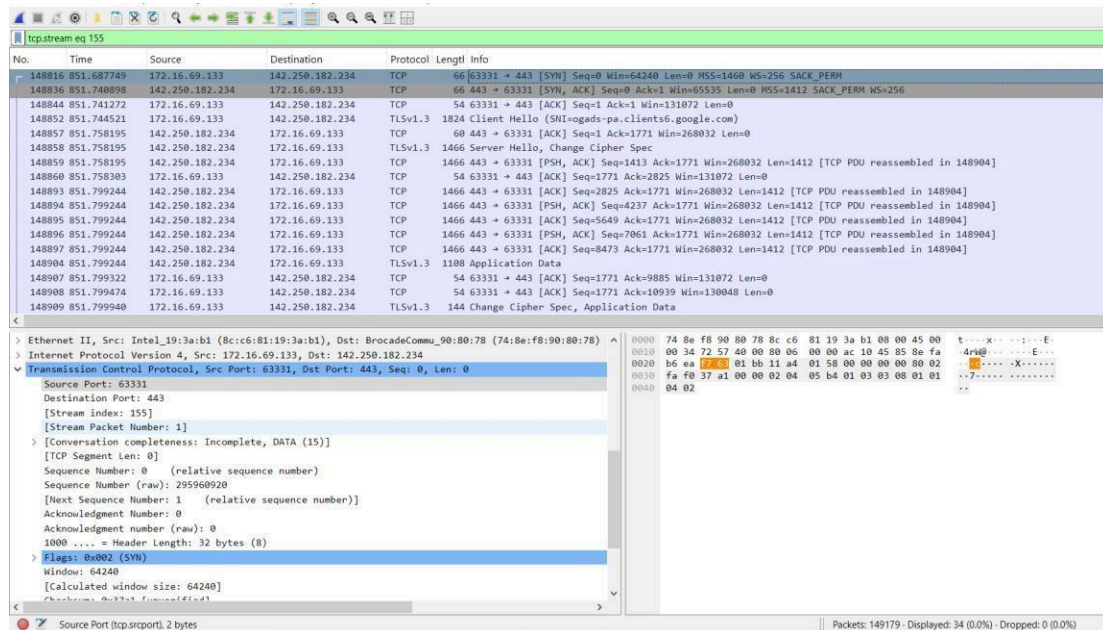
Observation and Analysis of ICMP Packet 85695 -

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, and Internet Control Message Protocol
- Source IP Address: 192.168.1.9
- Destination IP Address: 142.250.192.206
- Packet Length: 74 Bytes

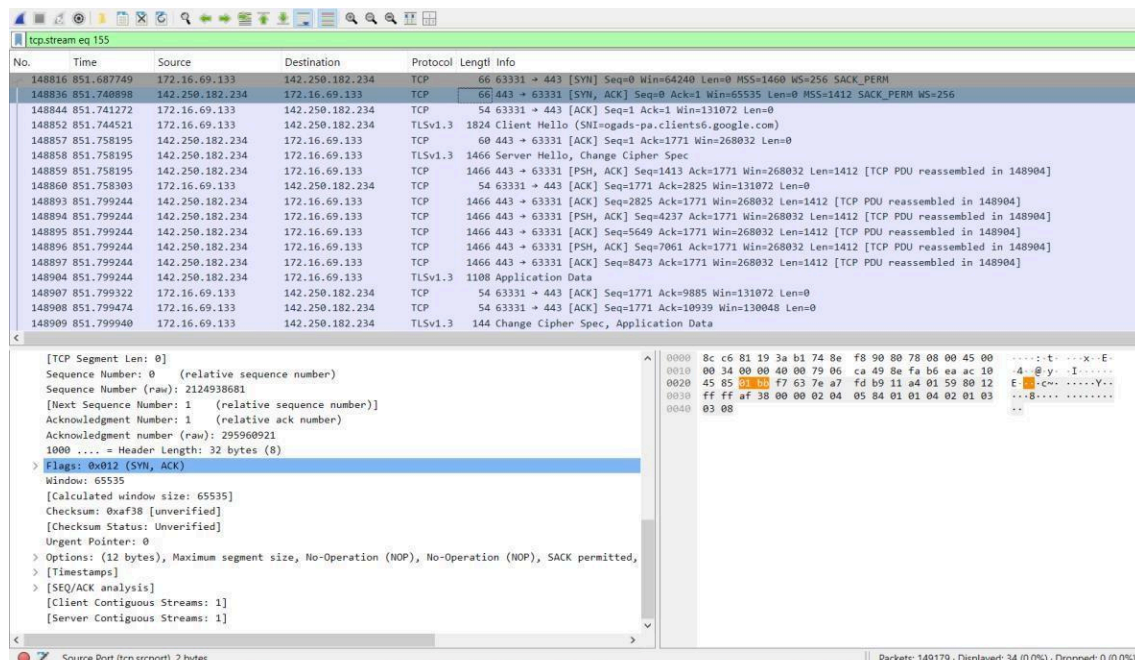
Observation and Analysis of ICMP Packet 85696 -

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, and Internet Control Message Protocol
- Source IP Address: 142.250.192.206
- Destination IP Address: 192.168.1.9
- Packet Length: 74 Bytes

2. To analyze TCP packets and observe the TCP three-way handshake



Screenshot showing TCP Packet (SYN)



Screenshot showing TCP Packet (SYN, ACK)

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.stream eq 155

No.	Time	Source	Destination	Protocol	Length	Info
148816	851.687749	172.16.69.133	142.250.182.234	TCP	66	63331 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
148836	851.740898	142.250.182.234	172.16.69.133	TCP	66	443 → 63331 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM WS=256
148844	851.741272	172.16.69.133	142.250.182.234	TCP	54	63331 → 443 [ACK] Seq=1 Ack=1 Win=131072 Len=0
148852	851.744521	172.16.69.133	142.250.182.234	TLSv1.3	1824	Client Hello (SHI=ogads-pa.clients6.google.com)
148857	851.758195	142.250.182.234	172.16.69.133	TCP	60	443 → 63331 [ACK] Seq=1 Ack=1771 Win=268032 Len=0
148858	851.758195	142.250.182.234	172.16.69.133	TLSv1.3	1466	Server Hello, Change Cipher Spec
148859	851.758195	142.250.182.234	172.16.69.133	TCP	1466	443 → 63331 [PSH, ACK] Seq=1413 Ack=1771 Win=268032 Len=1412 [TCP PDU reassembled in 148904]
148860	851.758303	172.16.69.133	142.250.182.234	TCP	54	63331 → 443 [ACK] Seq=1771 Ack=2825 Win=131072 Len=0
148893	851.799244	142.250.182.234	172.16.69.133	TCP	1466	443 → 63331 [ACK] Seq=2825 Ack=1771 Win=268032 Len=1412 [TCP PDU reassembled in 148904]
148894	851.799244	142.250.182.234	172.16.69.133	TCP	1466	443 → 63331 [PSH, ACK] Seq=4237 Ack=1771 Win=268032 Len=1412 [TCP PDU reassembled in 148904]
148895	851.799244	142.250.182.234	172.16.69.133	TCP	1466	443 → 63331 [ACK] Seq=5649 Ack=1771 Win=268032 Len=1412 [TCP PDU reassembled in 148904]
148896	851.799244	142.250.182.234	172.16.69.133	TCP	1466	443 → 63331 [PSH, ACK] Seq=7061 Ack=1771 Win=268032 Len=1412 [TCP PDU reassembled in 148904]
148897	851.799244	142.250.182.234	172.16.69.133	TCP	1466	443 → 63331 [ACK] Seq=8473 Ack=1771 Win=268032 Len=1412 [TCP PDU reassembled in 148904]
148904	851.799244	142.250.182.234	172.16.69.133	TLSv1.3	1108	Application Data
148907	851.799322	172.16.69.133	142.250.182.234	TCP	54	63331 → 443 [ACK] Seq=1771 Ack=9885 Win=131072 Len=0
148908	851.799474	172.16.69.133	142.250.182.234	TCP	54	63331 → 443 [ACK] Seq=1771 Ack=10939 Win=130048 Len=0
148909	851.799940	172.16.69.133	142.250.182.234	TLSv1.3	144	Change Cipher Spec, Application Data

[Stream Packet Number: 3]
[Conversation completeness: Incomplete, DATA (15)]
[TCP Segment Len: 0]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 295960921
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 2124938682
0101 = Header Length: 20 bytes (5)
Flags: 0x010 (ACK)
Window: 512
[Calculated window size: 131072]
[Window size scaling factor: 256]
Checksum: 0x3795 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
[Timestamps]
[SEQ/ACK analysis]
Followed by: ...

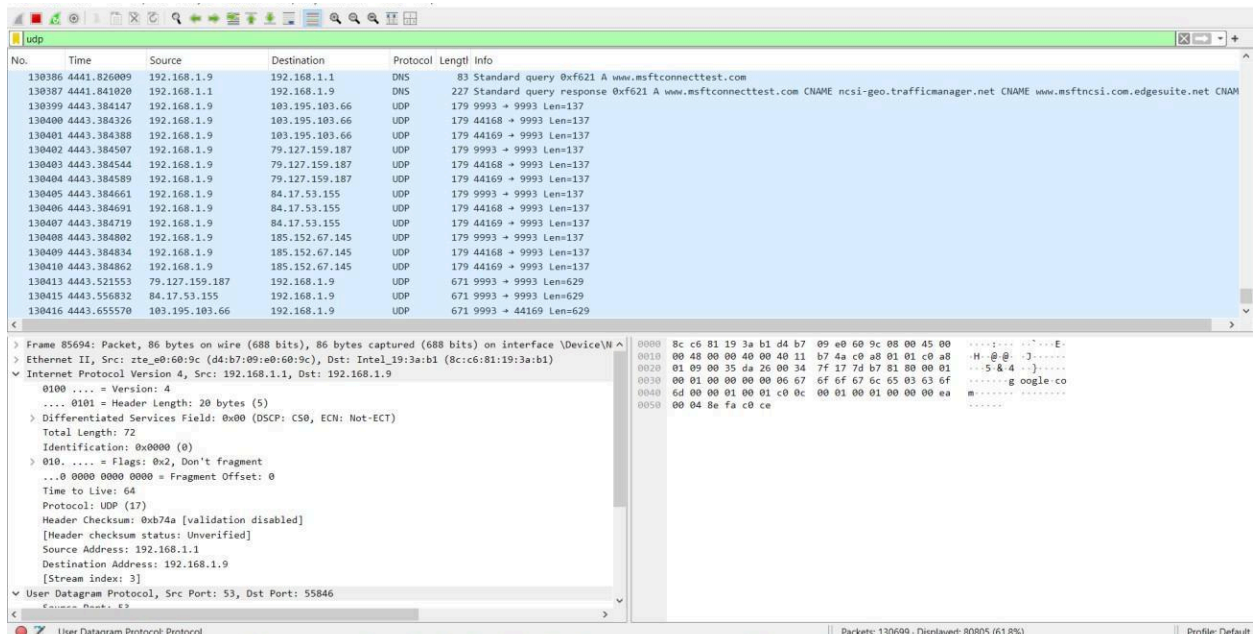
0000 74 8e f8 90 80 78 8c c6 81 19 3a b1 08 00 45 00
0010 00 28 72 5e 40 00 80 06 00 00 ac 10 45 85 8e fa
0020 b6 ea f7 63 01 bb 11 a4 01 59 7e a7 fd ba 50 10
0030 02 00 37 95 00 00

Flags (tcp.flags), 12 bits

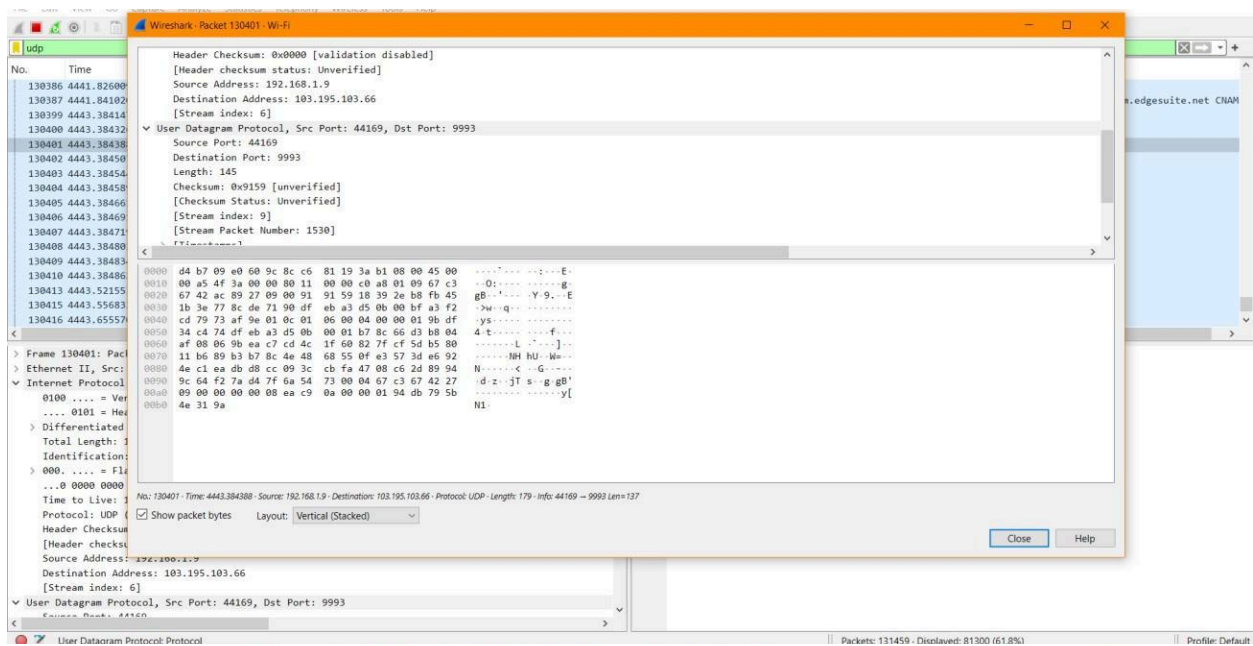
Packets: 149179 · Displayed: 34 (0.0%) · Dropped: 0 (0.0%)

Screenshot showing TCP Packet (ACK)

3. To capture and analyze UDP packets



Screenshot showing UDP Packets Capture

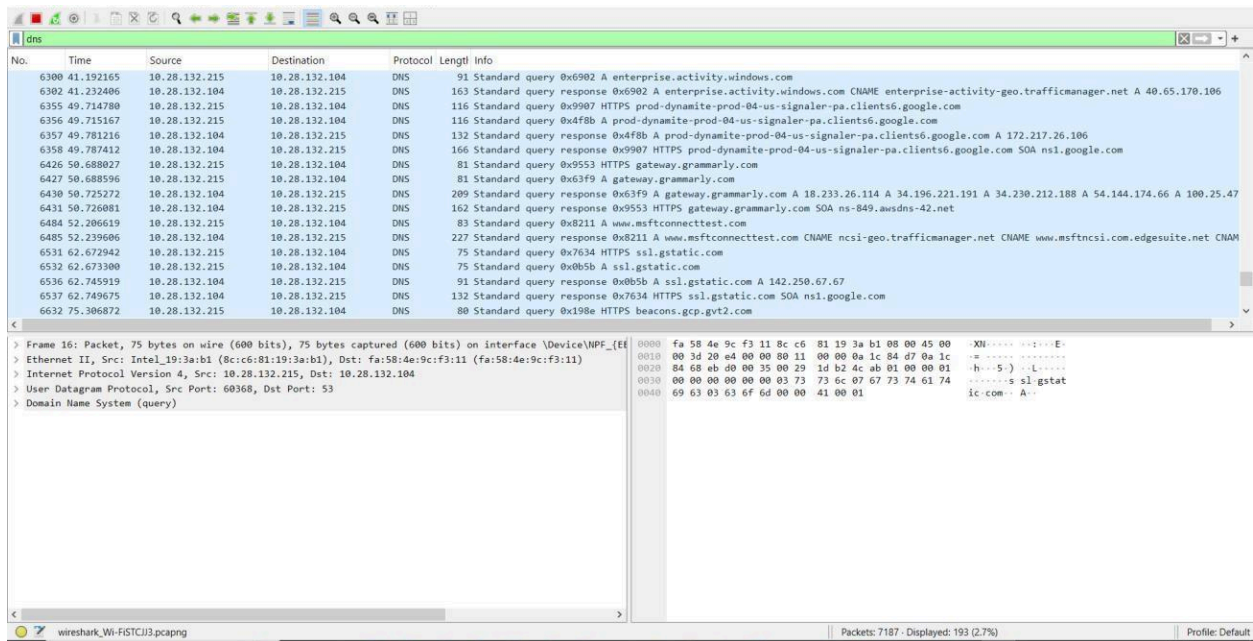


Screenshot showing details of a UDP Packet

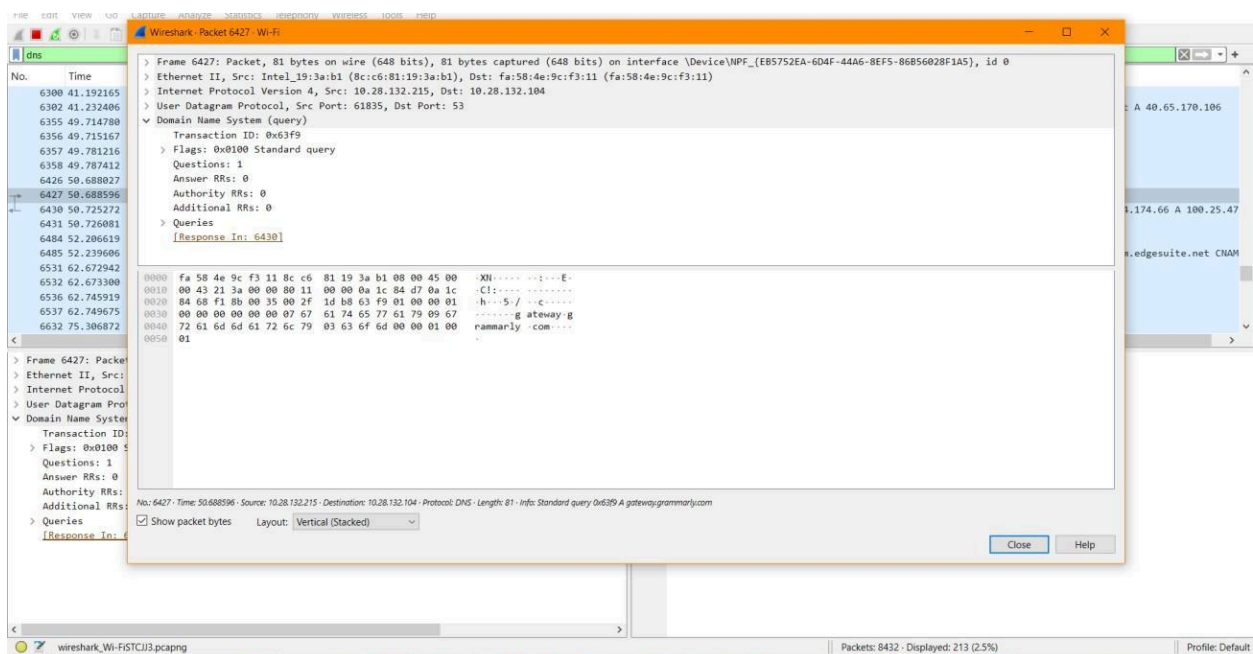
Observation: UDP Packet 130401 –

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, and User Datagram Protocol
- Source IP Address: 192.168.1.9
- Destination IP Address: 103.195.103.66
- Packet Length: 179 Bytes

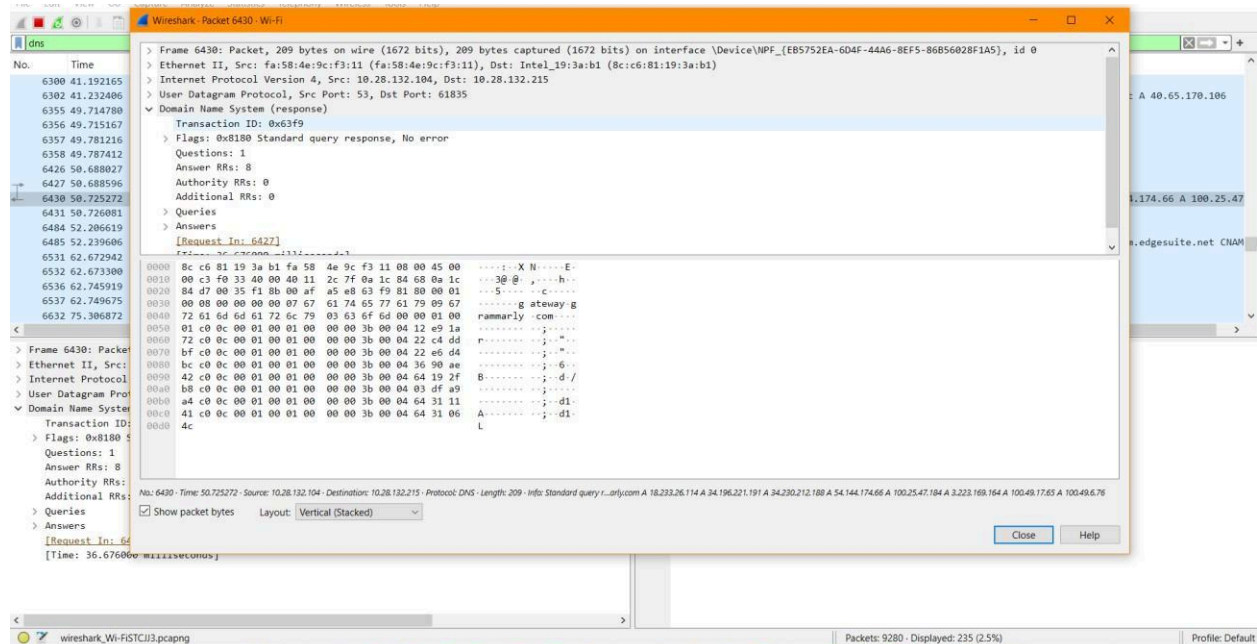
4. To capture and analyze DNS query and response packets



Screenshot showing DNS Packets Capture



Screenshot showing details of a DNS Packet 6427 (Standard query request)



Screenshot showing details of a DNS Packet 6430 (Standard query response)

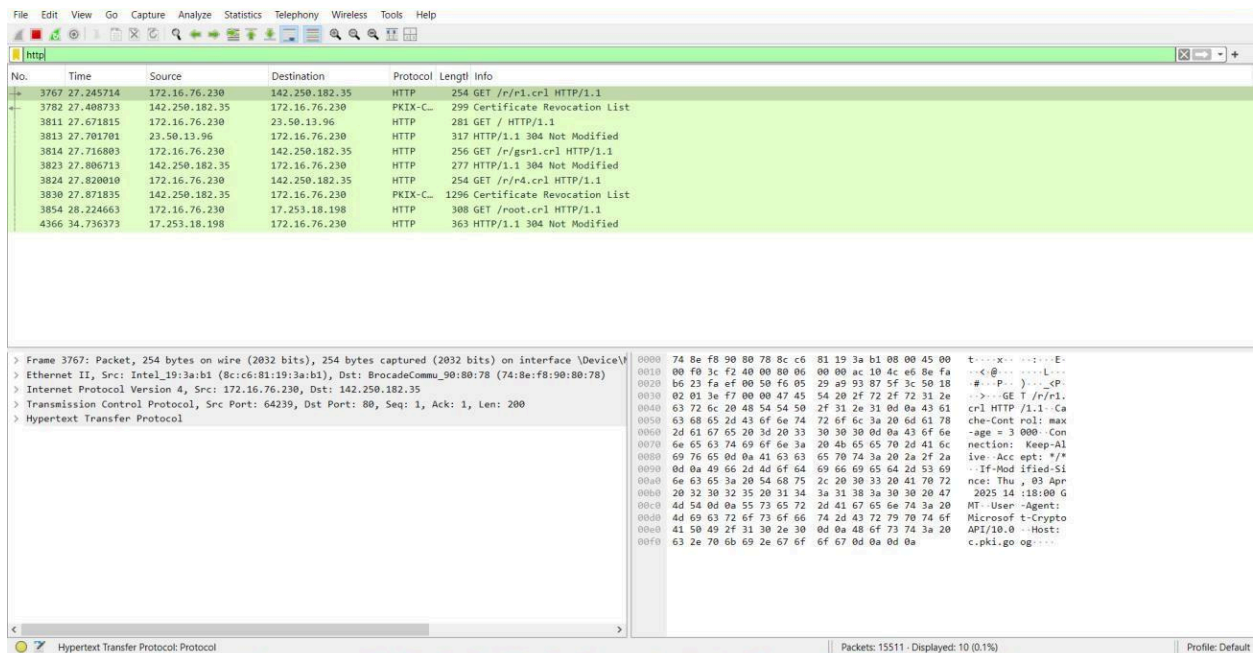
Observation: DNS Packet 6427 –

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, and User Datagram Protocol
- Source IP Address: 10.28.132.215
- Destination IP Address: 10.28.132.104
- Packet Length: 81 Bytes
- Queries - gateway.grammarly.com: type A, class IN

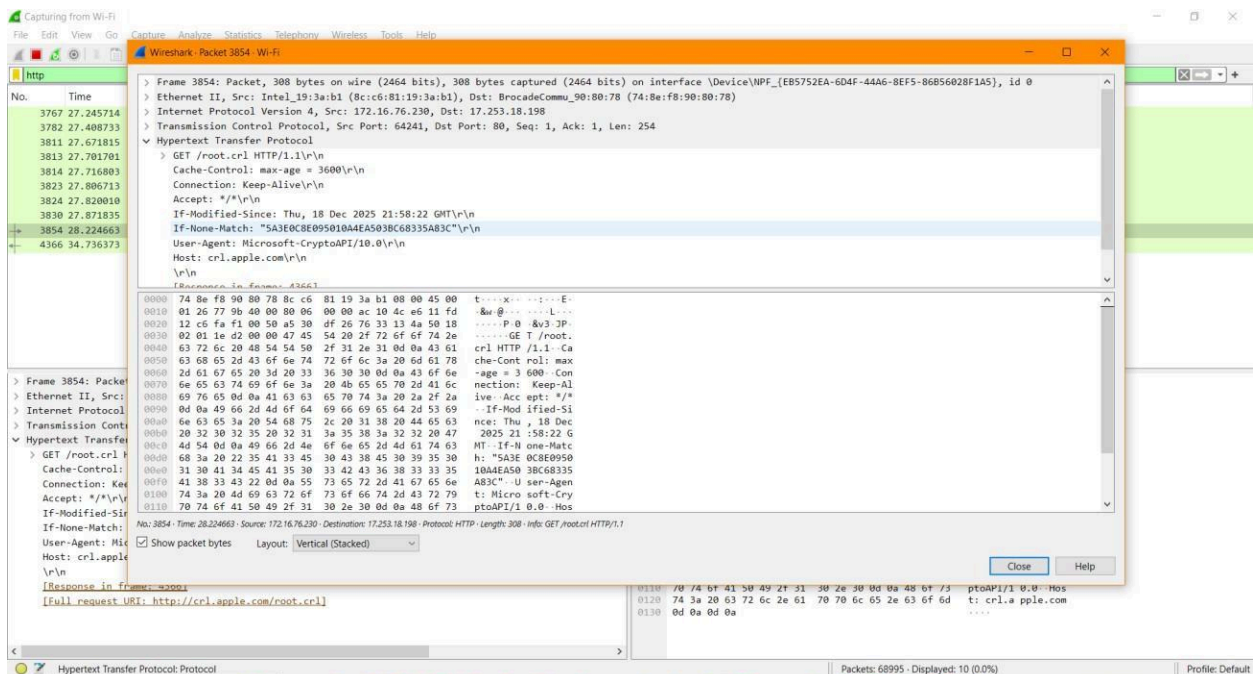
Observation: DNS Packet 6430 –

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, and User Datagram Protocol
- Source IP Address: 10.28.132.104
- Destination IP Address: 10.28.132.215
- Packet Length: 209 Bytes

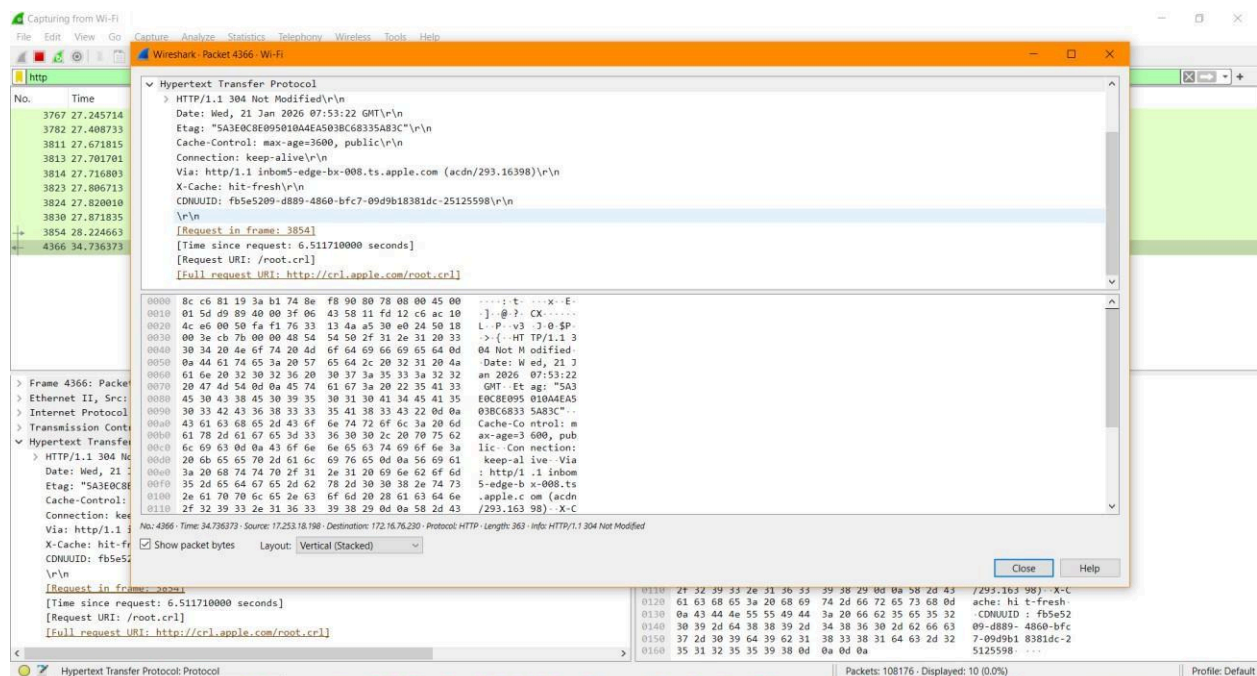
5. To capture and analyze HTTP packets



Screenshot showing HTTP Packets Capture



Screenshot showing details of a HTTP Packet (HTTP Request)



Screenshot showing details of a HTTP Packet (HTTP Response)

Observation: HTTP Request 3854 –

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, Transmission Control Protocol, Hypertext Transfer Protocol
- Source IP Address: 172.16.76.230
- Destination IP Address: 172.253.18.198
- Packet Length: 308

Bytes Observation: HTTP Response

4366 –

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, Transmission Control Protocol, Hypertext Transfer Protocol
- Source IP Address: 172.253.18.198
- Destination IP Address: 172.16.76.230
- Packet Length: 363 Bytes

6. To analyze ARP request and reply packets

The screenshot displays the Wireshark network protocol analyzer interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. The toolbar contains icons for various functions like opening files, saving, and analyzing. The packet list pane shows a list of captured packets, with the first packet being an ARP request from 172.16.74.134 to 172.16.64.1. The packet details pane shows the structure of Frame 21, which is an ARP request. The packet bytes pane shows the raw data of the packet, including the Ethernet II header, Internet Protocol Version 4 header, and ARP data.

No.	Time	Source	Destination	Protocol	Length	Info
4	0.103477	BrocadeCommu_90:80::...	Broadcast	ARP	60	Who has 172.16.74.134? Tell 172.16.64.1
21	0.196499	BrocadeCommu_90:80::...	Broadcast	ARP	60	Who has 172.16.73.115? Tell 172.16.64.1
22	0.206316	BrocadeCommu_90:80::...	Broadcast	ARP	60	Who has 172.16.77.107? Tell 172.16.64.1
23	0.206316	BrocadeCommu_90:80::...	Broadcast	ARP	60	Who has 172.16.93.88? Tell 172.16.64.1
24	0.206316	BrocadeCommu_90:80::...	Broadcast	ARP	60	Who has 172.16.75.224? Tell 172.16.64.1
25	0.206316	BrocadeCommu_90:80::...	Broadcast	ARP	60	Who has 172.16.95.46? Tell 172.16.64.1
26	0.206316	BrocadeCommu_90:80::...	Broadcast	ARP	60	Who has 172.16.68.235? Tell 172.16.64.1
27	0.206316	BrocadeCommu_90:80::...	Broadcast	ARP	60	Who has 172.16.74.134? Tell 172.16.64.1
28	0.206316	BrocadeCommu_90:80::...	Broadcast	ARP	60	Who has 172.16.73.129? Tell 172.16.64.1
29	0.206316	BrocadeCommu_90:80::...	Broadcast	ARP	60	Who has 172.16.69.106? Tell 172.16.64.1
30	0.206316	BrocadeCommu_90:80::...	Broadcast	ARP	60	Who has 172.16.74.102? Tell 172.16.64.1
31	0.206316	BrocadeCommu_90:80::...	Broadcast	ARP	60	Who has 172.16.91.60? Tell 172.16.64.1
32	0.206316	BrocadeCommu_90:80::...	Broadcast	ARP	60	Who has 172.16.81.154? Tell 172.16.64.1
33	0.206316	BrocadeCommu_90:80::...	Broadcast	ARP	60	Who has 172.16.92.219? Tell 172.16.64.1
34	0.206316	BrocadeCommu_90:80::...	Broadcast	ARP	60	Who has 172.16.94.96? Tell 172.16.64.1
35	0.206316	BrocadeCommu_90:80::...	Broadcast	ARP	60	Who has 172.16.91.186? Tell 172.16.64.1
36	0.206316	BrocadeCommu_90:80::...	Broadcast	ARP	60	Who has 172.16.83.209? Tell 172.16.64.1

Frame 21: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface \Device\NPF...
Section number: 1
> Interface id: 0 (\Device\NPF_{EB5752EA-6D4F-44A6-8EF5-86B56028F1A5})
Encapsulation type: Ethernet (1)
Arrival Time: Apr 1, 2026 13:25:54.747861000 India Standard Time
UTC Arrival Time: Apr 1, 2026 07:55:54.747861000 UTC
Epoch Arrival Time: 1775030154.747861000
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 91.281000 milliseconds]
[Time delta from previous displayed frame: 93.022000 milliseconds]
[Time since reference or first frame: 196.499000 milliseconds]
Frame Number: 21
Frame Length: 60 bytes (480 bits)
Capture Length: 60 bytes (480 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:arp]
Character encoding: ASCII (0)
[Collapse Rule Name: ARP]

Address Resolution Protocol: Protocol

Packets: 149179 - Displayed: 38629 (25.9%) - Dropped: 0 (0.0%)

Screenshot showing ARP Packets Capture

7. To identify source and destination MAC and IP addresses in captured packets

The screenshot displays the Wireshark network protocol analyzer interface. At the top, a menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. Below the menu is a toolbar with various icons for packet capture and analysis. The main window is divided into three panes:

- Packet List Pane:** Shows a list of captured packets. The first five packets are HTTP GET requests from 172.16.69.133 to 34.223.124.45. The sixth packet is an OCSP response. The seventh and eighth packets are HTTP GET requests from 172.16.69.133 to 104.18.27.120.
- Packet Details Pane:** Shows the hierarchical structure of the selected packet (Stream index: 99). It includes Internet Protocol Version 4, Hypertext Transfer Protocol, and Transmission Control Protocol details.
- Packet Bytes Pane:** Shows the raw data of the selected packet in hexadecimal and ASCII.

The selected packet (Stream index: 99) is an HTTP GET request from 172.16.69.133 to 34.223.124.45. The details pane shows the following information:

- Internet Protocol Version 4, Src: 172.16.69.133, Dst: 34.223.124.45
- 0100 = Version: 4
- 0101 = Header Length: 20 bytes (5)
- > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
- Total Length: 467
- Identification: 0x14da (5338)
- > 010. = Flags: 0x2, Don't fragment
- ...0 0000 0000 0000 = Fragment Offset: 0
- Time to Live: 128
- Protocol: TCP (6)
- Header Checksum: 0x0000 [validation disabled]
- [Header checksum status: Unverified]
- Source Address: 172.16.69.133
- Destination Address: 34.223.124.45
- [Stream index: 790]
- > Transmission Control Protocol, Src Port: 63964, Dst Port: 80, Seq: 1, Ack: 1, Len: 427
- > Hypertext Transfer Protocol

The packet bytes pane shows the raw data of the selected packet in hexadecimal and ASCII. The status bar at the bottom indicates that 365675 packets were captured, 6 (0.0%) were displayed, and 0 (0.0%) were dropped.

Screenshot showing Source and Destination IP Address of a HTTP

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help						
http						
No.	Time	Source	Destination	Protocol	Length	Info
38328	14.136535	172.16.69.133	34.223.124.45	HTTP	481	GET / HTTP/1.1
80726	27.793747	172.16.69.133	34.223.124.45	HTTP	507	GET / HTTP/1.1
321033	122.067090	172.16.69.133	23.11.39.161	HTTP	288	GET /MFewTzBNMEswSTA3BgUrDgMcGgUABBTcrjrydRytK2BApF3GSPypFHBxR5XtQQUs9tIpPmhxdIuNkHMEWNPYimBS8YCEAI5PUjXakJaf
321464	122.188176	23.11.39.161	172.16.69.133	OCSP	538	Response
350216	134.149918	172.16.69.133	104.18.27.120	HTTP	129	GET / HTTP/1.1
351492	134.773127	104.18.27.120	172.16.69.133	HTTP	61	HTTP/1.1 200 OK (text/html)

> Frame 38328: Packet, 481 bytes on wire (3848 bits), 481 bytes captured (3848 bits) on interface \Device\NPF{...}		0000	74 8e f8 90 80 70 8c c6	81 19 3a b1 08 00 45 00	...
Ethernet II, Src: Intel_19:3a:b1 (8c:c6:81:19:3a:b1), Dst: BrocadeCommu_90:80:78 (74:8e:f8:90:80:78)		0010	01 d3 14 da 40 00 80 06	00 00 ac 10 45 85 22 df	...
> Destination: BrocadeCommu_90:80:78 (74:8e:f8:90:80:78)		0020	7c 2d f9 dc 00 50 ad c6	eb 9d 2c d0 1e 6e 50 18	...
> Source: Intel_19:3a:b1 (8c:c6:81:19:3a:b1)		0030	02 01 92 67 00 00 47 45	54 20 2f d0 48 54 54 50	...
Type: IPv4 (0x0800)		0040	2f 31 2e 31 0d 0a 48 6f	73 74 3a 20 6e 65 76 65	...
[Stream index: 99]		0050	72 73 73 6c 2e 63 6f 6d	8d 0a 43 6f 6e 6e 65 63	...
> Internet Protocol Version 4, Src: 172.16.69.133, Dst: 34.223.124.45		0060	74 69 6f 6e 3a 20 6b 65	65 70 2d 61 6c 69 76 65	...
> Transmission Control Protocol, Src Port: 63964, Dst Port: 80, Seq: 1, Ack: 1, Len: 427		0070	0d 0a 55 70 67 72 61 64	65 2d 49 6e 73 65 63 75	...
> Hypertext Transfer Protocol		0080	72 65 2d 52 65 71 75 65	73 74 73 3a 20 31 0d 0a	...
		0090	55 73 65 72 2d 41 67 65	6e 74 3a 20 4d 6f 7a 69	...
		00a0	6c 6c 61 2f 35 2e 30 20	28 57 69 6e 64 6f 77 73	...
		00b0	20 4e 54 20 31 30 2e 30	3b 20 57 69 6e 36 34 3b	...
		00c0	20 78 36 34 29 20 41 70	70 6c 65 57 65 62 4b 69	...
		00d0	74 2f 35 33 37 2e 33 36	20 2b 4b 48 54 4d 4c 2c	...
		00e0	20 6c 69 6b 65 20 47 65	63 6b 6f 29 20 43 68 72	...
		00f0	6f 6d 65 2f 31 34 36 2e	30 2e 30 2e 30 20 53 61	...
		0100	66 61 72 69 2f 35 33 37	2e 33 36 0d 0a 41 63 63	...
		0110	65 70 74 3a 20 74 65 78	74 2f 68 74 6d 6c 2c 61	...
		0120	70 70 6c 69 63 61 74 69	6f 6e 2f 78 68 74 6d 6c	...
		0130	2b 78 6d 6c 2c 61 70 78	6c 69 63 61 74 69 6f 6e	...
		0140	2f 78 6d 6c 3b 71 3d 30	2e 39 2c 69 6d 61 67 65	...
		0150	2f 61 76 69 66 2c 69 6d	61 67 65 2f 77 65 62 70	...
		0160	2c 69 6d 61 67 65 2f 61	70 6e 67 2c 2a 2f 2a 3b	...

Destination Hardware Address (eth.dst), 6 bytes

Packets: 365675 - Displayed: 6 (0.0%) - Dropped: 0 (0.0%)

Screenshot showing Source and Destination MAC Address of a HTTP Packet

Observation: HTTP Packet 38328 –

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, Transmission Control Protocol, Hypertext Transfer Protocol
- Source IP Address: 172.16.69.133
- Destination IP Address: 34.223.124.45
- Source MAC Address: 8c:c6:81:19:3a:b1
- Destination MAC Address: 74:8e:f8:90:80:78
- Packet Length: 481 Bytes